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Roll No.: _____

Amrita Vishwa Vidyapeetham

B.Sc(Hons)/Int. M.Sc. Degree Examination – June 2024

Second Semester

Mathematics & Physics

22MAT113 Multivariable Calculus

Time : Three hours

Maximum : 100 Marks

Answer all questions.

Qn No.	Questions	Marks	CO	Bloom's Taxonomy level
1	Find the limits. (a) $\lim_{(x,y)\to(4,3)}\left(\frac{\sqrt{x}-\sqrt{y+1}}{x-y-1}\right)$. (b) $\lim_{(x,y)\to(0,0)}\left(\frac{x\sin y}{x^2+1}\right)$	3	CO2	L3
2	Let $f(x,y)=\tan^{-1}(y/x)$, find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.	3	CO2	L2
3	Assume that the equation $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=1$ defines z as a function of x and y. Find the values of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at the point $(2,3,6)$.	3	CO2	L3
4	Let $z=4e^x \ln y$ where $x=\ln(u\cos v)$ and $y=u\sin v$. Find $\frac{\partial z}{\partial v}$ at $(2,\pi/4)$.	3	CO2	L2
5	Find a parametrization for the portion of the paraboloid $z=x^2+y^2-4$, $z\le 0$.	3	CO1	L2
6	Find the directional derivative of the function, $f(x,y)=\ln(x^2+y^2)$, at the point $(3,0)$ in the direction of $\vec{a}=\hat{\imath}-\hat{\jmath}$.	5	CO1	L5
7	Find a parametric equation for the line tangent to the curve of intersection of the surfaces $x+y^2+2z=4$ and $x=1$ at the point $(1,1,1)$.	5	CO3	L4

8	Evaluate $\int_C f(x,y)\,ds$, where $f(x,y) = x^2 - y$ and C is the portion of the circle $x^2 + y^2 = 4$ in the first quadrant from $(0,2)$ to $(\sqrt{2},\sqrt{2})$.	5	CO2	L4
9	Find the work done by $\vec{F} = 2y\,\hat{\imath} + 3x\,\hat{\jmath} + (x+y)\,\hat{k}$ over the curve $C: \vec{r}(t) = (\cos t)\hat{\imath} + (\sin t)\hat{\jmath} + \left({}^{t}/_{6}\right)\hat{k},\ 0 \le t \le 2\pi$, in the direction of increasing t.	5	CO3	L3
10	Find the area of the region cut from the plane $x + 2y + 2z = 5$ by the cylinder whose walls are $x = y^2$ and $x = 2 - y^2$.	5	CO4	L5
11	Find equations for the tangent plane and normal line to the surface: $\cos \pi x - x^2 y + e^{xz} + yz = 4$ at the point $P_0(1,-1,3)$.	8	CO2	L3
12	Use a parametrization to express the area of the portion of the cylinder $x^2 + z^2 = 10$ between the planes $y = -1$ and $y = 1$ a double integral and then evaluate the integral.	8	CO5	L2
13	Show that curl $(\vec{F}) = 0$ for $\vec{F} = 3x^2\hat{\imath} + \frac{z^2}{y}\hat{\jmath} + 2z \ln y\,\hat{k}$ and hence evaluate $\int_{(1,1,1)}^{(1,2,3)} 3x^2 dx + \frac{z^2}{y} dy + 2z \ln y dz$.	8	CO5	L3
14	Evaluate the surface Integral $\iint_S G(x,y,z)\,d\sigma$, where $G(x,y,z) = y + z$ and S is the surface of the wedge in the first octant bounded by the coordinate planes and the planes $x = 2$ and $y + z = 1$.	8	CO1	L2
15	Find the flux $\iint_S \vec{F}.\hat{n}\,d\sigma$ across the surface through the portion of the cone $z = \sqrt{x^2 + y^2}$ between the planes $z = 1$ and $z = 2$, for $\vec{F} = -x\hat{\imath} - y\hat{\jmath} + z^2\hat{k}$ outward.	8	CO3	L2
16	State Green's theorem and hence verify it for $\vec{F} = (x^2 + y^2)\,\hat{\imath} - 2xy\,\hat{\jmath}$ taken round the rectangle bounded by the lines $x = \pm 1, y = 0, y = 2$.	10	CO3	L5
17	Verify Stoke's theorem for $\vec{F} = [y^2, 0, x^2]$ and the surface S is the paraboloid $z = 1 - x^2 - y^2,\ z \ge 0$.	10	CO5	L4

L_1-Remember, L_2-Understand, L_3-Apply, L_4-Analyze, L_5-Evaluate, L_6-Create